Zane (Chih-Hsuan) Tsai

CONTACT

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EDUCATION

Master of Science Mechanical Engineering

National Taiwan University
2018 Sep. – 2021 Jun. Taipei, Taiwan

Bachelor of Science Mechanical Engineering

National Taiwan University
2013 Sep. – 2017 Jun. Taipei, Taiwan

PUBLICATION

Development of an immersive SLAM-based VR system for teleoperation of a mobile manipulator in an unknown environment

C.Y. Kuo, C.C. Huang, *C.H. Tsai*, Y.S. Shi, S. Smith, Computers in Industry, Volume 132, 2021

SKILLS

C++ C# Python Claude Code
ROS/ROS2 Linux Git
Qt6 WPF WinForm
JIRA Confluence
AutoCAD SolidWorks

LANGUAGES

Mandarin — Native
English — Fluent (TOEIC 900)
Japanese — Proficient (JLPT N1)

EXPEIRENCE

Software Design Engineer

Delta Electronics Inc. 2023 Nov. – 2025 Sep. Taipei, Taiwan

- Implemented a CV-based visual alignment system using stereo vision and eye-in-hand calibration, significantly improving feeding precision for automated screw-driving and dispensing.
- Refactored 3+ core communication layer components (Socket, Modbus, SerialPort) using Factory Pattern, enhancing system scalability and maintainability.
- Developed hardware control modules and API integrations for 5+ devices, including Delta motion control cards, robot arms, conveyors, electric screwdrivers, laser displacement sensors, torque meters, etc.
- Deployed software for 3+ equipment types (screw-driving, dispensing, labeling) on-site, troubleshooting in communication diagnostics, hardware control, vision calibration, and manufacturing flow validation.
- Authored user manuals and developer documentation for reusable modules to facilitate secondary development and facilitate client integration.

HMI Software Engineer

Syntec Technology Inc. 2021 Nov. – 2023 Jan. Hsinchu, Taiwan

- Developed teaching pendant HMI (WinForm) for Articulated, Gantry, and SCARA robots; collaborated with PMs to deliver 5+ major features and resolve 10+ critical bugs, including UI logic and design, PLC integration, register specification, and command editor CRUD, etc.
- Integrated a 2D vision tracking suite by implementing 9-point calibration and frame transformation, enabling real-time workpiece detection and handling.
- Optimized automated testing by maintaining Batch scripts and log analysis tools across 5+ software versions, increasing QA efficiency and identifying 5+ critical bugs prior to release..
- Participated in Taipei Automation Expo and authored a competitive analysis and potential-demand report on palletizing functionality.

PROJECT

Simple Control (Side Project) 2025

- Designed a hardware-agnostic simulation platform using C++ and Qt based on SOLID principles and Design Patterns, enable component modularity and customizable manufacturing workflows.

Dynamic SLAM (Master's Thesis) 2021

- Proposed a mask-based SLAM front-end module for filtering dynamic objects and projecting 3D point cloud using RealSense RGB-D camera; validated on TUM dynamic-object datasets using ROS (C++/Python), achieving ~80% reduction in trajectory error.

Inchworm Bio-Robot (Bachelor's Project) 2017

- Engineered and simulated motion planning for an inchworm robot in MATLAB using Inverse Kinematics, successfully designing 2 gait patterns for stepping and climbing.